



CHE 230 Molecular Systems in Chemical Engineering

Spring 2026

Class Time: M/W 8:00 – 9:15 PM CST

Format: In-person EIB Room 271

Instructor Information

Instructor

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(Roshan)

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Office Location & Hours

EIB Room 270

By appointment (Email)

Teaching Assistant

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Office Location & Hours

EIB/Zoom

TBD

Course Description

Prerequisites: CHEM 232.

The objective of this course is to provide a general background on **molecular systems** for undergraduate level students. Fundamental topics that will be introduced include chemical bonding; synthesis mechanisms; nanoparticle nucleation; molecular characterization; the mechanical, electrical, and optical properties of molecular systems; crystal structure and defects; diffusion and Brownian motion; interfacial phenomena; phase diagrams of molecular systems; and applications of synthetic macromolecules, biomacromolecules and nanomaterials. This course covers examples of molecular systems used in energy, medicine, environmental and other technology applications.

Recommended References:

References will be posted online on Blackboard

- Materials Science and Engineering: An Introduction, Willam D. Callister Jr and David G. Rethwisch, Wiley; 9th edition (December 2013)

Expectations and Goals

ABET Learning Outcomes

1. Ability to apply knowledge of mathematics, science, and engineering to solve material problems related to chemical engineering.
2. Development of an understanding of professional and ethical responsibility.
3. Broad education is necessary to understand the impact of engineering and scientific solutions in a global, economic, environmental, and societal context.

Course Learning Outcomes

One of the main objectives of the course is to familiarize the students with the fundamental concepts of Materials Science and Engineering which will be used as background knowledge for the understanding of specialized problems in the field of Chemical Engineering. At the end, students will be able to:

1. Understand the fundamental and essential concepts used in materials chemistry and molecular systems
2. Classify the molecular systems, nanomaterials, and materials systems.
3. Know the basic methods to synthesize and characterize chemicals and materials systems
4. Understand the basic properties that characterize the behavior of molecular systems
5. Select appropriate type of material for specific application
6. Offer different approaches to modify structure/microstructure to get desired properties

Assessment Methods

1. Homework assignment will assess student's mastery of the basic principles and concepts of molecular systems.
2. Exams and quizzes will assess students' ability to design, characterize, and evaluate polymers.
3. Individual case studies will assess the students' ability to identify important problems and discuss the challenges and possible solutions.

Graded Items	% Of Final Grade
Homework	30% (6% x 5)
Pop Quizzes	20% (4% x 5)
Mid Term Exams (2)	30% (2 x 15%)
Case Study	20%

Homework: There will be ~5 homework assignments.

Quizzes: 5 quizzes, each is 4%. Get half credit just for being present for the quiz

Midterm exam: There will be two exams.

Final report: At the end of the course, you need to do a one-page study on a given topic.

Late/Missed Work

1. Late Homework Policy: Homework can be turned in 1 lecture period late without excuse for up to 50% credit.

Tentative Course Schedule

Week	Day	Topic	Assignments	
Week 1	01/12	Introduction to Molecular Systems	HW1	
	01/14			
Week 2	01/19	MLK Day. No Classes.		
	01/21	Molecular Structure and Defects		
Week 3	01/26			Diffusion & Material Properties from molecular structure
	01/28			
Week 4	02/02	Molecular Interactions: Influence on Macroscopic Properties		
	02/04			
Week 5	02/09		HW2	
	02/11			
Week 6	02/16			HW3
	02/18			
Week 7	02/23	HW4		
	02/25			
Week 8	03/02		Exam 1	
	03/04		Engineering Functional Materials	
Week 9	03/09			HW5
	03/11			
Week 10	03/16	Spring Break		
	03/18			
Week 11	03/23			
	03/25			
Week 12	03/30		Nanoscale Systems	
	04/01			
Week 13	04/06	Sustainability & Circular Materials		
	04/08			
Week 14	04/13			Commercializing Products with Molecular Engineering
	04/15			
Week 15	04/20		Exam 2 Review	
	04/22			
Week 16	04/27	Special Topics		
	04/29	Special Topics		

Course Materials

Reading materials will be assigned for each topic before instruction and lecture slides will be assigned after each class. Please monitor Blackboard for updates.

Our Blackboard course site has been organized to facilitate both access to course materials and to a broad array of UIC campus resources and community information such as advising support, online events, health, and wellness tools, etc. The information is provided at the university, college, and department/course levels to help you more efficiently navigate the resources and tools you need to be successful in your courses this semester. Students are expected to log into the course site regularly to learn about any developments related to the course as well as to

upload assignments and communicate with classmates. For all technical questions about Blackboard, email the CATE Learning Technology Solutions team LTS@uic.edu.

Please protect the integrity of all course materials and content. By enrolling in this course, you agree to honor this request. Be mindful of the hard work and time that instructors and TAs put into creating course materials such as exam and quiz questions, worksheets, lecture videos, and reading materials. Please do not upload course materials not created by you onto third-party websites or share content with anyone who is not enrolled in our course. I am grateful for your cooperation in honoring this important request.

Course Policies

Academic Integrity

UIC is an academic community committed to providing an environment in which research, learning, and scholarship can flourish and in which all endeavors are guided by academic and professional integrity. In this community, all members including faculty, administrators, staff, and students alike share the responsibility to uphold the highest standards of academic honesty and quality of academic work so that such a collegial and productive environment exists.

As a student and member of the UIC community, you are expected to adhere to the Community Standards of integrity, accountability, and respect in all your academic endeavors. When accusations of academic dishonesty occur, the Office of the Dean of Students investigates and adjudicates suspected violations of this student code. Unacceptable behavior includes cheating, unauthorized collaboration, fabrication or falsification, plagiarism, multiple submissions without instructor permission, using unauthorized study aids, coercion regarding grading or evaluation of coursework, and facilitating academic misconduct. Please review the UIC Student Disciplinary Policy for additional information about the process by which instances of academic misconduct are handled towards the goal of developing responsible student behavior.

By submitting your assignments for grading you acknowledge these terms, you declare that your work is solely your own, and you promise that, unless authorized by the instructor or proctor, you have not communicated with anyone in any way during an exam or other online assessment. Let's embrace what it means to be a UIC community member and together be committed to the values of integrity.

<https://dos.uic.edu/community-standards/>

<https://dos.uic.edu/community-standards/academic-integrity/>

COVID 19

Please review UIC updates for guidance <https://today.uic.edu/summer-covid-19-guidance>

Excused Absence

Students should speak to their faculty regarding their absence first. When possible, your instructors should be notified of your absence, prior to missing class. It is your responsibility to connect with your faculty about any coursework you missed during your absence and coordinate a plan moving forward. Documentation is always required. Requests will not be accepted without appropriate documents. Please note that documentation must include the specific dates of absence from campus

Please review the UIC page for Emergencies and Absences for more information on examples of approved scenarios <https://dos.uic.edu/student-assistance/academic-concerns/emergencies-and-absences/>

Writing standards

Please proof your papers and e-mail messages before submitting them. Grades for content, completeness, organization, spelling, grammar, and punctuation, as well as demonstration of knowledge gained in the course and your ability to apply it.

Attendance / Participation Policy

Attendance is mandatory and highly recommended for the success of this class. Please email me if you face an unexpected situation that may impede your attendance, participation in required class and exam sessions, or timely completion of assignments.

Regrading

Any request for a regrade must be made within one week of the assignment being returned to you. If you think there has been a simple arithmetic error on your assignment then write a note explaining the error, attach this to the front of the assignment and turn it in to your TA. If the error is confirmed, then the points will be added to your score before the end of the semester. If you think there was an error in grading that is not an arithmetic error, write a brief note explaining why you think more points should be awarded, attach this to the front of the assignment and turn it into your TA. If you do turn in a regrade request, you should make a photocopy for your own records. All regrades (except arithmetic errors) will be done at the end of the semester after final grades are calculated. If the points you request affect your final grade, then I will reevaluate the assignment for the contested points.

Religious Holidays

I will make every effort to avoid scheduling exams or requiring student projects to be submitted on religious holidays. If you wish to observe your religious holidays, please notify me by the tenth day of the semester of the date when you will be absent unless the religious holiday is observed on or before the tenth day of the semester. In such cases, please notify me at least five days in advance of the date when you will be absent. I will make every reasonable effort to honor your request and not penalize you for missing the class. If an examination or project is due during your absence, you will be given an exam or assignment equivalent to the one completed by those students in attendance. Students may appeal through campus grievance procedures for religious accommodations.

Our Inclusive Learning Environment

UIC values diversity and inclusion. Regardless of age, disability, ethnicity, race, gender, gender identity, sexual orientation, socioeconomic status, geographic background, religion, political ideology, language, or culture, we expect all members of this class to contribute to a respectful, welcoming, and inclusive environment for every other member of our class. If there are aspects of the instruction or design of this course that result in barriers to your inclusion, engagement, accurate assessment, or achievement, please notify me as soon as possible.

If your name does not match the name on my class roster, please let me know as soon as possible. My pronouns are [she/her](#). I welcome your pronouns if you would like to share them with me. For more information about pronouns, see this page: <https://www.mypronouns.org/what-and-why>.

Disability Accommodation

UIC is committed to full inclusion and participation of people with disabilities in all aspects of university life. If you face or anticipate disability-related barriers while at UIC, please connect with the Disability Resource Center (DRC) at drc.uic.edu, via email at drc@uic.edu, or call (312) 413-2183 to create a plan for reasonable accommodations. To receive accommodations, you will need to disclose the disability to the DRC, complete an interactive registration process with the DRC, and provide me with a Letter of Accommodation (LOA). Upon receipt of a LOA, I will gladly work with you and the DRC to implement approved accommodations.

Classroom Etiquette

Community Agreement: Ground Rules for a Safe/Brave Space	Community Agreement: Online Etiquette
Be present (turn off cell phones and remove yourself from other distractions) Be respectful Assume good will	Be in a quiet place and use headphones Mute microphone unless talking Video on when speaking or engaging in group activities
Challenge with care -approach discussion as a “think out loud”	Quit all other screens (email, text, social media) to create a classroom “presence” (also helps with internet stability)
Take space/make space Identities stay, learning leaves	Actively participate in class Raise hand if you have a question to ask verbally
Try not to make assumptions, seek to understand, not to judge Be open to challenges as an opportunity to learn something new Be open to different perspectives Debate the concepts not the person	Use Chat feature if you prefer to type your question No side conversations
Be flexible when things don’t work Share helpful tips	Be willing to work together Be mindful of one another’s privacy --do not invite outsiders into our classroom Use preferred names and gender pronouns

Tips for Success

Time commitment

Since this is a 3-credit hour course, you should plan to spend 3 hours per week online/in-person and an additional 6 to 9 hours on readings, research, discussions, assignments, etc. The rule-of-thumb is 2 to 3 hours per week outside the “classroom” for every credit hour per week in the classroom. If you do the math, it adds to **9–12 hours per week, total**.

There are many ways for you to show us what you know or can do and how you are learning, through your effort, interactions, and class participation, application of critical thinking skills to solve real world problems, and performance on assignments and exams. This class has been structured to help all students get the support and guidance needed to succeed in your learning. The following tips will help you be successful in this class and in other classes throughout your career at UIC.

- Ask questions during class
- Complete assignments on time
- Participate in class discussions
- Individually review your notes after class
- Post questions on the discussion board or ask questions during drop-in (office) hours about concepts or procedures that seemed confusing
- Attend / engage in drop-in (office) hours with me and your TA
- Work in groups with your classmates both during class and outside of class

If we are online components, please consider these:

- Have practically unlimited access to computer and internet services
- Commit to the specified amount of time {average hrs./week} to the course
- Be self-motivated and self-disciplined, keeping up with assignments and completing coursework on time
- Communicate through writing as well as use video and audio methods as needed for oral communication
- Think through ideas before responding
- Accept critical thinking and decision making as part of the learning process
- Be open minded about sharing life, work, and educational experiences as part of the learning process
- Speak up if problems arise because non-verbal communication not possible in an online setting
- Embrace a mindset in which you believe high quality learning can take place without going to a traditional classroom

Resources

UIC [Tutoring Resources](#)

Writing Center <https://writingcenter.uic.edu/>

Math and Science Learning Center <https://mslc.uic.edu/>

UIC Library <http://library.uic.edu/> <http://researchguides.uic.edu/>

Academic Center for Excellence (ACE) <https://ace.uic.edu/>

[First-at-LAS](#) Academic Success Program, focusing on LAS first-generation students.

[Offices](#) supporting the UIC Undergraduate Experience and Academic Programs.

Counseling Services <https://counseling.uic.edu/>

Campus Advocacy Network <http://can.uic.edu/>

[Student Guide for Information Technology](#)

UIC's Virtual Private Network (VPN) services

<https://acc.uic.edu/services/infrastructure/network/virtual-private-network/>

Tech Solution Loaner Laptops

<https://help.uillinois.edu/TDClient/37/uic/Requests/ServiceDet?ID=450>

Campus Security

You may request an escort to accompany you to your campus destination on foot by calling (312)996-2830, and between 11:00 pm and 7:00 am you can dial the Red Car service (312)996-6800 if you are alone and need to leave the building

You can download the UIC SAFE app, a free personal security tool for students, faculty, and staff. It allows you easy contact with dispatchers and first responders in case of emergency. Navigate to the UIC SAFE Toolkit to download the app: <https://ready.uic.edu/digital-materials/uic-safe-app/>.

By dialing 5-5555 from a campus phone, you can summon Police or Fire for any on-campus emergency. You may also set up the complete number, (312)355-5555, on speed-dial on your cell phone.

Disclaimer

This syllabus is intended to give you guidance on what may be covered during the semester and will be followed as closely as possible. However, as the instructor, I reserve the right to modify, supplement and make changes as the course needs arise. I will communicate such changes in advance as in-class announcements and in writing via Blackboard Announcements.